

TOUCHLESS VOLUME CONTROL USING HAND GESTURES

Abstract

Touchless volume control using hand gestures is a human–computer interaction system that enables users to adjust audio levels without physical contact. The system uses a standard webcam to capture real-time video, which is processed using computer vision techniques to detect and track hand landmarks. The distance between specific fingers is calculated and mapped to system volume levels, allowing smooth and intuitive control. The system is implemented using Python, OpenCV, and MediaPipe, ensuring accuracy, responsiveness, and cost-effectiveness. It eliminates the need for additional hardware, improves hygiene and accessibility, and demonstrates the effectiveness of gesture-based interaction in modern smart environments.

Objectives

- Develop a real-time system to control volume using hand gestures.
- Detect and track hand landmarks accurately with computer vision.
- Adjust volume smoothly based on finger movements.
- Improve accessibility and hygiene through touchless interaction

Methodology

The methodology of the touchless volume control system enables real-time, contactless interaction by capturing live video through a webcam and processing each frame. OpenCV is used for preprocessing tasks such as resizing, color conversion, and noise reduction to improve detection accuracy. MediaPipe is utilized to detect and track hand landmarks, including fingertips and joints, which are analyzed to recognize gestures. The distance between the thumb and index finger is calculated and mapped to predefined system volume levels, allowing smooth and intuitive volume adjustment. Python libraries interface with the operating system to control audio output directly, while visual feedback such as tracking lines and volume bars guides the user. This lightweight and cost-effective system requires no additional hardware and demonstrates an efficient approach to modern touchless human–computer interaction.

Conclusions

- Provides a touchless and intuitive way to control system volume using hand gestures.
- Enables real-time, smooth, and accurate volume adjustment without physical contact.
- Enhances usability, accessibility, and hygiene, making it suitable for smart environments.